# Advanced HTML

ਪੂਰੇ ਨਾਮ + ਕੰਮ + ਸੌਖੀ ਵਿਆਖਿਆ + Code Examples

| | |
|---|---|
| **ਕਿਸ ਲਈ?** | HTML basics ਤੋਂ ਅੱਗੇ ਜਾਣ ਵਾਲੇ student ਲਈ |
| **Style** | Punjabi + English terms, beginner-friendly |
| **Topics** | Semantic HTML, Forms, Input Types, Tables, Images, Audio/Video, iframe, Links, SEO, Accessibility, data-*, Special Elements, Entities, Head/Meta, Best Practices |

Tip: ਹਰ example ਨੂੰ Acode/VS Code ਵਿੱਚ copy ਕਰਕੇ browser ਵਿੱਚ run ਕਰੋ।

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**Learning order:** Semantic → Forms → Inputs → Tables → Media → iframe → Links → SEO → Accessibility → data-* → Special elements → best practices.

# 1. Semantic HTML

**Full name:** Semantic HyperText Markup Language elements

**Kam:** Page ਦੇ ਹਿੱਸੇ ਨੂੰ ਉਸਦੇ meaning ਅਨੁਸਾਰ tag ਦੇਣਾ। SEO, accessibility ਅਤੇ clean structure ਲਈ useful.

## Example

```
<header>
  <h1>My Student Hub</h1>
</header>
<nav>
  <a href="index.html">Home</a>
</nav>
<main>
  <section>
    <h2>10th Class</h2>
    <p>Study notes here.</p>
  </section>
</main>
<footer>© 2026 Student Hub</footer>
```

> **Important:** ਹਰ ਚੀਜ਼ ਲਈ ਸਿਰਫ  ਨਾ ਵਰਤੋ। Meaning ਵਾਲਾ tag ਹੋਵੇ ਤਾਂ semantic tag ਵਰਤੋ।

# 2. Advanced Forms

**Full name:** HTML Form elements

**Kam:** User ਤੋਂ data ਲੈਣਾ: login, search, contact, registration ਆਦਿ।

## Example

```
<form action="/submit" method="post">
  <label for="name">Name</label>
  <input id="name" name="name" type="text">

  <label for="class">Class</label>
  <select id="class" name="class">
    <option>10th</option>
    <option>11th</option>
  </select>

  <textarea name="message" placeholder="Message"></textarea>
  <button type="submit">Send</button>
</form>
```

> **Important:** label ਨੂੰ input ਨਾਲ `for` + `id` ਰਾਹੀਂ connect ਕਰਨਾ ਚੰਗੀ practice ਹੈ।

# 3. Input Types

**Full name:** HTML input `type` attribute

**Kam:** ਇੱਕੋ `<input>` tag ਨੂੰ ਵੱਖ-ਵੱਖ ਕਿਸਮਾਂ ਦਾ field ਬਣਾਉਣਾ।

## Example

```
<input type="text">
<input type="email">
<input type="password">
<input type="number">
<input type="date">
<input type="file">
<input type="range" min="0" max="100">
<input type="color">
<input type="checkbox">
<input type="radio" name="gender">
```

> **Important:** type ਬਦਲਣ ਨਾਲ browser field ਦਾ behaviour ਅਤੇ built-in validation ਵੀ ਬਦਲ ਸਕਦਾ ਹੈ।

## Common types

| Type | Use |
|---|---|
| text | Normal text |
| email | Email address |
| password | Password input |
| number | Numeric input |
| date | Date picker |
| file | File selection |
| range | Slider |
| color | Color picker |

# 4. Form Validation Attributes

**Full name:** Built-in HTML form validation attributes

**Kam:** Submit ਤੋਂ ਪਹਿਲਾਂ basic rules ਲਗਾਉਣਾ।

## Example

```
<input type="email" required>
<input type="text" minlength="3" maxlength="20" required>
<input type="number" min="1" max="100">
<input type="text" pattern="[A-Za-z ]+">
<input autocomplete="email">
```

| Attribute | Kam |
|---|---|
| required | Field ਖਾਲੀ ਨਹੀਂ ਛੱਡ ਸਕਦੇ |
| minlength / maxlength | Text ਦੀ length control |
| min / max | Number range control |
| pattern | Specific pattern match |
| autocomplete | Browser ਨੂੰ autofill hint |

> **Important:** HTML validation user experience ਲਈ useful ਹੈ; real applications ਵਿੱਚ server-side validation ਵੀ ਜ਼ਰੂਰੀ ਹੋ ਸਕਦੀ ਹੈ।

# 5. Tables Advanced

**Full name:** HTML table structure elements

**Kam:** Rows ਅਤੇ columns ਵਿੱਚ data ਦਿਖਾਉਣਾ।

## Example

```
<table>
  <thead>
    <tr><th>Name</th><th>Marks</th></tr>
  </thead>
  <tbody>
    <tr><td>Nav</td><td>85</td></tr>
    <tr><td>Aman</td><td>90</td></tr>
  </tbody>
  <tfoot>
    <tr><td>Total</td><td>175</td></tr>
  </tfoot>
</table>
```

> **Important:** `colspan` ਕਈ columns ਨੂੰ join ਕਰਦਾ ਹੈ; `rowspan` ਕਈ rows ਨੂੰ join ਕਰਦਾ ਹੈ।

# 6. Images Advanced

**Full name:** Responsive image elements and attributes

**Kam:** ਵੱਖ screen sizes ਲਈ suitable image ਦੇਣਾ ਅਤੇ loading performance improve ਕਰਨਾ।

## Example

```
<img src="small.jpg"
     srcset="small.jpg 480w, large.jpg 1200w"
     sizes="(max-width: 600px) 480px, 1200px"
     alt="Students studying"
     loading="lazy">

<picture>
  <source media="(max-width: 600px)" srcset="mobile.jpg">
  <img src="desktop.jpg" alt="Study desk">
</picture>
```

> **Important:** `alt` accessibility ਲਈ important ਹੈ। `loading="lazy"` suitable images ਨੂੰ later load ਕਰਨ ਵਿੱਚ ਮਦਦ ਕਰ ਸਕਦਾ ਹੈ।

# 7. Audio & Video

**Full name:** HTML media elements

**Kam:** Website ਵਿੱਚ audio/video files embed ਕਰਨਾ।

## Example

```
<audio controls>
  <source src="song.mp3" type="audio/mpeg">
</audio>

<video controls width="640" poster="thumb.jpg">
  <source src="lesson.mp4" type="video/mp4">
  <track src="captions.vtt" kind="subtitles"
         srclang="en" label="English">
</video>
```

> **Important:** captions ਲਈ `track` useful ਹੈ, ਖਾਸ ਕਰਕੇ accessibility ਲਈ।

# 8. iframe

**Full name:** Inline Frame element

**Kam:** ਦੂਜੇ webpage/resource ਨੂੰ ਆਪਣੇ page ਦੇ ਅੰਦਰ embed ਕਰਨਾ।

## Example

```
<iframe
  src="https://example.com"
  title="Example website"
  width="600"
  height="400">
</iframe>
```

> **Important:** YouTube embeds, maps ਆਦਿ ਲਈ iframe ਵਰਤਿਆ ਜਾ ਸਕਦਾ ਹੈ। Meaningful `title` ਦੇਣਾ ਚੰਗੀ practice ਹੈ।

# 9. Links Advanced

**Full name:** Anchor element and link attributes

**Kam:** Pages, files ਜਾਂ resources ਨਾਲ navigation/connection ਬਣਾਉਣਾ।

## Example

```
<a href="about.html">About</a>
<a href="https://example.com" target="_blank" rel="noopener">
  Open site
</a>
<a href="notes.pdf" download>Download Notes</a>
```

| Attribute | Kam |
| --- | --- |
| href | Link destination |
| target | ਕਿੱਥੇ open ਹੋਵੇ |
| download | File download hint |
| rel | Relationship/security hint |

## 10. SEO & Meta

**Full name:** Search Engine Optimization and metadata

**Kam:** Browser ਅਤੇ search engines ਨੂੰ page ਬਾਰੇ useful information ਦੇਣਾ।

### Example

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
        content="width=device-width, initial-scale=1.0">
  <title>10th Class Study Hub</title>
  <meta name="description"
        content="Free study resources for students.">
</head>
```

> **Important:** Useful `title` ਅਤੇ description page ਨੂੰ better describe ਕਰਦੇ ਹਨ। Keyword stuffing ਤੋਂ ਬਚੋ।

# 11. Accessibility (A11y)

**Full name:** Web Accessibility

**Kam:** Website ਨੂੰ screen-reader, keyboard ਅਤੇ ਹੋਰ needs ਵਾਲੇ users ਲਈ usable ਬਣਾਉਣਾ।

## Example

```
<img src="teacher.jpg" alt="Teacher explaining a lesson">

<label for="email">Email</label>
<input id="email" type="email">

<button type="button">Save</button>
```

> **Important:** Semantic tags, alt text, labels ਅਤੇ keyboard-friendly controls accessibility better ਬਣਾਉਂਦੇ ਹਨ। ARIA ਨੂੰ ਜਿੱਥੇ native HTML ਕਾਫੀ ਹੈ ਉੱਥੇ unnecessarily ਨਾ ਵਰਤੋ।

# 12. data-* Attributes

**Full name:** Custom data attributes

**Kam:** Element ਨਾਲ custom data attach ਕਰਨਾ ਜਿਸਨੂੰ JavaScript read ਕਰ ਸਕਦੀ ਹੈ।

## Example

```
<button data-id="101" data-class="10th">Open</button>

<script>
const btn = document.querySelector('button');
console.log(btn.dataset.id);
console.log(btn.dataset.class);
</script>
```

> **Important:** ਤੁਸੀਂ `data-id`, `data-user`, `data-topic` ਵਰਗੇ custom names ਰੱਖ ਸਕਦੇ ਹੋ।

# 13. Special HTML Elements

**Full name:** Additional semantic / interactive HTML elements

**Kam:** Extra UI ਅਤੇ meaning ਵਾਲੇ built-in elements ਵਰਤਣਾ।

## Example

```
<details>
  <summary>HTML ਕੀ ਹੈ?</summary>
  <p>HTML webpage ਦਾ structure ਬਣਾਉਂਦੀ ਹੈ.</p>
</details>

<figure>
  <img src="book.jpg" alt="HTML book">
  <figcaption>HTML Study Book</figcaption>
</figure>

<dialog open>Welcome!</dialog>
```

| Element | Use |
| --- | --- |
| details + summary | Collapsible content |
| figure + figcaption | Media + caption |
| dialog | Dialog box / modal style UI |

# 14. HTML Entities

**Full name:** Character references / HTML entities

**Kam:** Reserved ਜਾਂ special characters ਨੂੰ text ਵਜੋਂ display ਕਰਨਾ।

## Examples

| Entity | Meaning |
|---|---|
| `<` | < |
| `>` | > |
| `&` | & |
| ` ` | Non-breaking space |
| `©` | © |

Example: `<h1>` ਲਿਖਣ ਨਾਲ browser ਇਸਨੂੰ tag ਦੀ ਥਾਂ text ਵਜੋਂ ਦਿਖਾ ਸਕਦਾ ਹੈ।

# 15. Advanced Head / Resources

**Full name:** Head metadata and external resources

**Kam:** CSS, favicon ਅਤੇ browser metadata connect/define ਕਰਨਾ।

## Example

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="icon" href="favicon.ico">
  <link rel="stylesheet" href="style.css">
</head>
```

> **Important:** viewport mobile responsive layout ਲਈ basic ਅਤੇ important ਹੈ।

# 16. Professional Best Practices

**Full name:** Clean, accessible and maintainable HTML practices

**Kam:** Code ਨੂੰ clean, readable ਅਤੇ future ਵਿੱਚ edit ਕਰਨ ਲਈ easy ਰੱਖਣਾ।

## Example structure

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub</title>
</head>
<body>
  <header>...</header>
  <main>...</main>
  <footer>...</footer>
</body>
</html>
```

> **Important:** Consistent indentation, meaningful names, semantic tags, valid nesting, useful alt text ਅਤੇ unnecessary tags ਤੋਂ ਬਚਣਾ professional HTML ਦੀਆਂ important habits ਹਨ।

## Quick Revision

| Thing | Purpose | Example |
|---|---|---|
| <header> | Page/section header | <header>...</header> |
| <nav> | Navigation links | <nav>...</nav> |
| required | Field must be filled | <input required> |
| pattern | Pattern validation | pattern="[A-Z]+" |
| srcset | Responsive image sources | srcset="a.jpg 480w, b.jpg 1200w" |
| target | Link target | target="_blank" |
| alt | Alternative image text | alt="Book" |
| data-* | Custom data | data-id="10" |

**Practice challenge:** ਇੱਕ Student Hub page ਬਣਾਓ ਜਿਸ ਵਿੱਚ header, nav, main, section, form, table, image, video ਅਤੇ footer ਹੋਵੇ। ਪਹਿਲਾਂ HTML complete ਕਰੋ, ਫਿਰ CSS ਲਗਾਓ।